

VISTA: A Test-Time Self-Improving Video Generation Agent

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Abstract

Despite rapid advances in text-to-video synthesis, generated video quality remains critically dependent on precise user prompts. Existing test-time optimization methods, successful in other domains, struggle with the multi-faceted nature of video. In this work, we introduce VISTA (Video Iterative Self-improvement Agent), a novel multi-agent system that autonomously improves video generation through refining prompts and self-evaluating generated videos in an iterative loop. VISTA first decomposes a user’s idea into a structured temporal plan. After generation, the best video is identified through a robust pairwise tournament. This winning video is then critiqued by a trio of specialized agents focusing on visual, audio, and contextual fidelity. Finally, a reasoning agent synthesizes this feedback to introspectively rewrite and enhance the prompt for the next generation cycle. Experiments on single- and multi-scene video generation scenarios show that while prior methods yield inconsistent gains over state-of-the-art (SOTA) models such as Veo 3, VISTA consistently improves video quality and alignment with user intent, achieving up to 60% pairwise win rate against prior methods. Human evaluators concur, preferring VISTA’s outputs in 66.4% of comparisons: <https://g-vista.github.io/>.

1. Introduction

Text-to-video (T2V) generation has seen significant progress, with models like Veo 3 [14] and Sora 2 [47] demonstrating a remarkable ability to generate coherent, high-quality video and audio from text. This progress positions T2V as a powerful tool for applications in storytelling, education, and content creation [19, 34, 40, 43, 57]. However, broader deployment is limited by several persistent challenges. Models often struggle with precise user [30, 32], consistent adherence to physics and commonsense [2, 25, 59], and a high sensitivity to the exact phrasing of input prompts [4, 8, 24]. This sensitivity, in particular, forces users into a laborious

trial-and-error cycle, requiring them to repeatedly tweak phrasing and filter outputs to achieve a desirable result.

Meanwhile, test-time optimization frameworks have shown promise in automatically improving generation quality and preference alignment for both text and image generation [15, 37, 44, 49, 52]. However, extending them to autonomous video generation presents substantially challenges. Unlike text or image data, videos unfold across multiple scenes, modalities, and high-level contextual meaning, making evaluation and optimization significantly more complex. Existing efforts often target only specific video properties, such as objects [8], harmless-accurate-helpful traits [4], or visual rewards [5, 23, 24, 31, 53]. Yet, to the best of our knowledge, no studies have unified visual, audio, and contextual quality in a single optimization framework, despite these together being key to user satisfaction.

We introduce VISTA (Video Iterative Self-improvement Agent), a novel multi-agent framework that self-improves video-audio generation at test time. Unlike existing agent studies [56, 58, 61] that focus on agentic workflows, VISTA introduces a different yet complementary capability of automated quality improvement through iterative, failure-focused optimization, jointly optimizing visual, audio, and context dimensions of videos. This process is guided by a comprehensive and configurable suite of evaluation metrics tailored to each, and is driven by four key components: (i) **Structured Video Prompt Planning** (Section 2.1), which transforms user input into temporally grounded, multi-scene, and multi-aspect descriptions; (ii) **Pairwise Tournament Selection** (Section 2.1), a probing-driven algorithm to identify the best video-prompt candidates; (iii) **Multi-Dimensional Multi-Agent Critiques** (Section 2.2), a triadic agent system that provides nuanced critiques motivated by the *Jury Decision Process* [26]; and a (iv) **Deep Thinking Prompting Agent**, which performs human-like introspective, structured reasoning to revise prompts in a targeted way. Experiments on a widely used single-scene benchmark and an internal multi-scene benchmark with rich instructions show that VISTA substantially outperforms prior test-time optimization methods, effectively self-corrects both clear and nuanced errors in generated videos, improving state-of-the-art (SOTA) T2V

*This work was done while Long was a Student Researcher at Google.

Direct Prompting (DP)



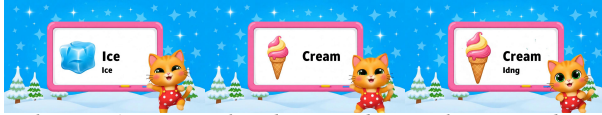
Single-scene [48]: The person's forehead creased with worry as he listened to bad news.



Single-scene [48]: A spaceship entering hyperdrive, stars streaking past as it accelerates.



Multi-scene-Interviews: The video features a man outdoors, asking a trivia question about a comedian known...



Multi-scene-Animation: The video is an educational animation designed for young children...

VISTA (Ours)



Table 1. Example videos generated by Veo 3 [14] with VISTA, showing improvements in both clear and subtle errors about visual fidelity, camera focus, smooth transitions, storylines, and sounds. Optimized prompts and more examples are in our project page.

models like Veo 3 by up to 60% under our metrics, further validated by human evaluations. Our work enables more reliable, user-aligned text-to-video generation and paves the way for broader video synthesis applications. This paper makes the following contributions:

- We propose VISTA, a multi-agent framework that emulates human-like prompt refinement to improve T2V generation. To our knowledge, VISTA is the first to self-improve the visual, audio, and context dimensions of videos jointly. See Table 1 for its generated videos.
- We meticulously develop VISTA’s components and their configurable evaluation criteria that enable fully autonomous, model-driven video evaluation and refinement.
- We conduct extensive experiments with in-depth analysis and human studies showing that VISTA consistently outperforms existing methods and improves user preferences.

2. VISTA

VISTA (Alg. 1 and Figure 5) is a modular, configurable framework for optimizing text-to-video generation. Given a user video prompt P , it produces an optimized video V^* and its refined prompt P^* through two phases: (i) **Initialization** and (ii) **Self-Improvement**, inspired by the human video optimization process via prompting. During (i), the prompt is parsed and planned into variants to generate candidate videos (Step 1), after which the best video-prompt pair is selected

(Step 2). In (ii), the system generates multi-dimensional, multi-agent critiques (Step 3), refines the prompt (Step 4), produces new videos, and reselects the champion pair (Step 2). This phase continues until a stopping criterion is met or the maximum number of iterations is reached.

2.1. Initialization Phase

Step 1. Structured Video Prompt Planning (Alg. 1-L1).

The user video generation prompt P is first parsed into m **timed sequences of scenes** as prompt candidates where each candidate is defined as $P_i := [S_{i,1}, S_{i,2}, \dots]$. By default, each scene configuration consists of nine properties spanning context, visual, and audio dimensions: (1) **Duration**: The length (in seconds) of the scene; (2) **Scene Type**: The type of the scene (e.g., montage); (3) **Characters**: Any key figures or entities on screen, whether human, animal, or object, that drive the scene (e.g., a flower); (4) **Actions**: Any deliberate and meaningful movements or behaviors that the characters above perform (e.g., the flower is blooming); (5) **Dialogues**: Any spoken lines, voiceovers, or on-screen text that drive the scene; (6) **Visual Environment**: Any environmental backdrop or setting that establishes the scene’s world (e.g., boundless sky canopy); (7) **Camera**: Any cinematographic technique, including framing, movement, or angle, that shapes the scene (e.g., a close-up capturing subtle emotion); (8) **Sounds**: Any auditory elements, such as soundtrack, voiceovers, or ambient noises, that enrich the scene (e.g., whispering wind); (9) **Moods**: Any prevailing tone or atmosphere that powers the scene (e.g., serene).

Algorithm 1 VISTA

Require: User prompt P , text-to-video model $T2V()$, multimodal MLLM(), iterations T , optional early-stop m , configurable criteria $\mathcal{M}_{user}^S, \mathcal{M}_{user}^C$

Ensure: Optimized video V^* and prompt P^*

[Initialization Phase]

- 1: $\mathcal{P} := \{P_1, \dots, P_m, P\} \leftarrow \text{PROMPTPLANNER}(P)$
- 2: $\mathcal{V} := \{V_1, \dots, V_n\}$ where $\{V_i, \dots\} \leftarrow T2V(P_j)$
- 3: $(V^*, P^*) \leftarrow \text{PAIRWISESELECT}(\mathcal{V}, \mathcal{P}, P, \mathcal{M}_{user}^S)$

[Self-Improvement Phase]

- 4: **for** $t = 1$ to T **do**
 - 5: $\mathcal{F}^t \leftarrow \text{MMAC}(V^*, P^*, P, \mathcal{M}_{user}^C)$
 - 6: $\mathcal{P}^t := \{P_1^t, \dots, P_m^t, P^*\} \leftarrow \text{PROMPTOPT}(P^*, P, \mathcal{F}^t)$
 - 7: $\mathcal{V}^t := \{V_1^t, \dots, V_n^t, V^*\}$ where $\{V_i^t, \dots\} \leftarrow T2V(P_j^t)$
 - 8: $(V^*, P^*) \leftarrow \text{PAIRWISESELECT}(\mathcal{V}^t, \mathcal{P}^t, P, \mathcal{M}_{user}^S)$
 - 9: **if** (V^*, P^*) unchanged for m iterations **then**
 - 10: **break**
 - 11: **end if**
 - 12: **end for**
 - 13: **return** (V^*, P^*)
-

VISTA implements a configurable interface over the above properties and planning constraints on scenes (see below). This introduces two main novelties over prior work [12, 22, 23, 48]: (i) **temporal scene-level decomposition**, where user prompts are structured as semantically coherent segments to support reasoning over complex content, and (ii) **fine-grained multi-modal video prompting**, enabling automated multi-dimensional critiques and self-improvement. A multi-modal LLM (MLLM) is used to obtain $S_{i,j}$ and, when needed, infer missing scene properties. To preserve fidelity to the user’s intent and avoid scenario drift, the system by default enforces planning constraints on *Realism*, *Relevancy*, and *Creativity*, though they are configurable and optional: videos are grounded in real-world physics unless the prompt specifies otherwise (e.g., animated or fantastical); only elements explicitly stated or implied are included, avoiding unnecessary invention; ambient sounds and visual and transition effects are encouraged when beneficial. See Supp.-Section 8.1.1 for defaults. We retain a residual P in $\mathcal{P}$ to support models that do not benefit from decomposition.

Step 2. Binary Tournament Video Selection with Probing Critiques (Alg. 1-L3, L10). After obtaining candidate video-prompt pairs, this step selects the pair (V^*, P^*) with the highest video quality for self-improvement. Conventional video evaluation methods often rely on complex metric-based systems [22], which are computationally expensive in practice [62]. Inspired by recent advances demonstrating the strong video understanding capabilities of MLLMs [7, 29], we employ an MLLM-as-a-Judge to evaluate videos on customizable evaluation criteria $\mathcal{M}_{user}^S$. However, scoring videos without ground-truth references is inherently sub-

jective and unreliable. We adopt a video pairwise comparison strategy, which better aligns with human preferences in reinforcement learning and language tasks, while avoiding model-induced biases [27, 35]. As detailed in Alg. 2, we iteratively reduce the set of candidate videos via **Binary Tournaments** [41]. At each iteration, videos are grouped into pairs and compared bidirectionally via swapping (Alg. 2-L5, L6); only the winning videos advance to the next round. We swap the videos to avoid token biases [63].

Comparing (V_i, V_j) , however, remains unreliable to align with human preferences. We employ a criteria-based approach [28] where the model judges videos across metrics $\mathcal{M}_{user}^S$. We meticulously design a default configuration where $\mathcal{M}_{user}^S := \{\text{Visual Fidelity; Physical Common-sense; Text-Video Alignment; Audio-Video Alignment; Engagement}\}$ capturing the core aspects of video quality, and the video winning on more criteria is the winner. While this approach is more principled, we still observe that model often fails to provide sufficiently critical assessments. We attribute this to the dual burden placed on the model: analyzing videos and comparing them simultaneously. To mitigate this, we introduce a two-step decomposition: first, for each video, the model generates probing critiques ($\mathcal{Q}$ in Alg. 2-L1) on $\mathcal{M}_{user}^S$, then these critiques are used to support the comparisons. Finally, we implement customizable penalty mechanisms for T2V failures during selection: by default, VISTA penalizes common failures in $\mathcal{M}_{user}^S$ unless the user prompt explicitly specifies otherwise. The final scores of video candidates V_i and V_j are s_i and s_j :

$$\begin{aligned} s_i &= \frac{1}{k} \sum_{C \in \mathcal{M}_{user}^S} [\delta(C, V_i, V_j) - \lambda \mathbb{1}(C, V_i)], \\ s_j &= \frac{1}{k} \sum_{C \in \mathcal{M}_{user}^S} [1 - \delta(C, V_i, V_j) - \lambda \mathbb{1}(C, V_j)] \end{aligned} \quad (1)$$

where $\delta(C, V_i, V_j) \in \{0, 0.5, 1\}$ represents the outcome of V_i against V_j on C , corresponding to $\{\text{Loss, Tie, Win}\}$, and $\mathbb{1}(C, V) \in \{0, 1\}$ indicates whether V violates C to some extent. The term λ is a penalty applied for violations. Noteworthy, $\mathbb{1}(C, V)$ can be customized to any preferred constraints, not necessarily restricted to $\mathcal{M}_{user}^C$. See Supp.-Section 8.2 for our default prompts and metrics definitions.

2.2. Self-Improvement Phase

Step 3. Multi-Dimensional Multi-Agent Critiques (MMAC; Alg. 1-L5). Given (P, V^*, P^*) , this step elicits targeted critiques to refine the prompt P^* . Video evaluation is difficult due to its multi-dimensional nature. We address this with a multi-agent critique framework decomposed into $\mathcal{D} = \{\text{Visual, Audio, Context}\}$, each assessed independently by a dedicated system. Evaluation criteria are configured through $\mathcal{M}_{user}^C$. While prior work [33, 65] proposed diverse metrics, recent SOTA models [14, 54] already

Algorithm 2 PAIRWISESELECT (Step 2)

Require: P ; video candidates $\mathcal{V} = \{V_1, \dots, V_n\}$ with prompts $\mathcal{P} = \{P_1, \dots, P_n\}$; optional user configurable $\mathcal{M}_{\text{user}}^S$
Ensure: Best video V^* and its prompt P^*

- 1: $\mathcal{Q} := \{Q_1, \dots, Q_n\} \leftarrow \text{MLLM}(\mathcal{V}, \mathcal{P}, \mathcal{M}_{\text{user}}^S)$
- 2: **while** $|\mathcal{V}| > 1$ **do**
- 3: Group $\mathcal{V}$ into pairs
- 4: **for each pair** (V_i, V_j) **do**
- 5: $(V_{\text{win}}^f, V_{\text{lose}}^f) \leftarrow \text{MLLM}(V_i, Q_i, V_j, Q_j, \mathcal{M}_{\text{user}}^S)$
- 6: $(V_{\text{win}}^s, V_{\text{lose}}^s) \leftarrow \text{MLLM}(V_j, Q_j, V_i, Q_i, \mathcal{M}_{\text{user}}^S)$
- 7: **if** $(V_{\text{win}}^f, V_{\text{lose}}^f) = (V_{\text{win}}^s, V_{\text{lose}}^s)$ **then**
- 8: $(V_{\text{win}}, V_{\text{lose}}) \leftarrow (V_{\text{win}}^f, V_{\text{lose}}^f)$
- 9: **else**
- 10: Randomly assign $(V_{\text{win}}, V_{\text{lose}})$
- 11: **end if**
- 12: $\mathcal{V}.\text{remove}(V_{\text{lose}})$; $\mathcal{P}.\text{remove}(P_{\text{lose}})$
- 13: **end for**
- 14: **end while**
- 15: **return** $(\mathcal{V}[0], \mathcal{P}[0])$

excel on most of them, suggesting they insufficiently differentiate between high-quality generations (see Section 4.1). Therefore, we carefully design a comprehensive default configuration for $\mathcal{M}_{\text{user}}^C$ that expose modality-specific failures even in SOTA T2V models. These are strategically selected and refined from [2, 3, 10, 33]:

- **Visual:** Visual Fidelity, Motions and Dynamics, Temporal Consistency, Camera Focus, Visual Safety.
- **Audio:** Audio Fidelity, Audio-Video Alignment, Audio Safety.
- **Context:** Situational Appropriateness, Semantic Coherence, Text-Video Alignment, Physical Commonsense, Engagement, Video Format (Beginning, Ending, Transitions).

See Supp.-Section 8.3 for the metrics’ definitions. These include both human-centric criteria such as situational appropriateness and video format, and fine-grained video metrics like visual fidelity¹. However, we observe that directly employing MLLM-as-a-Judge often yields shallow and unuseful critiques, even when explicitly prompted for criticism. This is because SOTA T2V models such as Veo 3 already produce high-quality outputs, pushing the critique task from identifying obvious failures to diagnosing not only them but also nuanced and complex deficiencies in visual, audio, and context. To implement a deeper, more diagnostic analysis, we introduce **Multi-Dimensional Multi-Agent Critiques (MMAC)**: inspired by the **Jury Decision Process** [26], for each evaluation dimension $D \in \mathcal{D}$, we construct a triadic

¹This set of metrics is more comprehensive than in Step 2 because this step prioritizes in-depth and granular critiques.

court consisting of a **Normal Judge** that critically assesses and scores the video on D ’s metrics in both good and bad faiths, an **Adversarial Judge** which generates probing questions, counterarguments, and scores to expose video flaws on D ’s metrics, and a **Meta Judge** that consolidates the judges from both sides:

$$\begin{aligned} \{C_D^+, S_D^+\} &\leftarrow J_D^+(P, V^*, P^*) && \text{(Normal Judge)} \\ \{C_D^-, S_D^-\} &\leftarrow J_D^-(P, V^*, P^*) && \text{(Adversarial Judge)} \\ \{C_D^*, S_D^*\} &\leftarrow J_D^*(P, C_D^+, S_D^+, C_D^-, S_D^-) && \text{(Meta Judge)} \end{aligned} \quad (2)$$

The MMAC is $\mathcal{F} := \{C_D^*, S_D^* | D \in \mathcal{D}\}$ where C_D and S_D are metrics’ critiques and scores (on a scale of 1-10, following Zheng et al. [64]).

Step 4. Prompt Optimization (Alg. 1-L6). After obtaining $\mathcal{F}$, this step refines P^* via a **Deep Thinking Prompting Agent (DTPA)**. Direct MLLM optimization often overcomplicates prompts and interprets critiques shallowly (see Section 4.3 for examples); DTPA instead performs a six-step, self-reflective reasoning to suggest prompt modifications in one chain-of-thought: (1) identifying video issues via metrics with low meta scores (≤ 8), (2) clarifying the expected outcome and success criteria, (3) evaluating the context sufficiency in the current prompt, (4) determining whether failures stem from model limitations or prompt, and (5) detecting potential conflicts or vagueness within the prompt itself. Based on this introspective analysis, the DTPA proposes a set of modification actions, which are then (6) reviewed and refined to ensure they fully address the failures identified in (1). See Supp.-Sections 8.5, 10.1 and 10.5 for VISTA’s prompts and its optimized prompts.

$$\mathcal{M} := \{M_1, \dots\} \leftarrow \text{DTPA}((P, P^*, \mathcal{F})), \quad (3)$$

where M_i are suggested modifications.

M_i are then used to sample improved prompts:

$$\mathcal{P} := \{P_1, \dots, P_n, P^*\} \leftarrow \text{MLLM}(P, P^*, \mathcal{M}) \quad (4)$$

3. Related Work

3.1. Text-To-Video Synthesis and Optimization

Recent years have seen major advances in T2V synthesis [14, 20, 46–48, 54], with SOTA models such as Sora and Veo 3 receiving widespread attention–Veo 3 notably pioneering high-quality audio-video generation. Yet, current models remain highly prompt-sensitive. Existing prompt optimization and test-time self-evolving (agentic) methods, while being successful in other domains [6, 9, 31, 36, 38, 51, 55], are limited in video optimization, often requiring white-box access or fine-tuning. For example, VideoAgent [53] refines video

plans by executing them online and using successful trajectories to fine-tune the generation model, while MotionPrompt [45] learns token embeddings to improve motion fidelity, and RAPO [8] rewrites and augments prompts but relies on target prompts during training. Inspired by test-time scaling with LLMs, Video-T1 [31] formulates test-time scaling for video generation as a search over denoising trajectories but does not consider further improving videos. Closest to our work, VPO [4] optimizes user inputs for harmlessness, accuracy, and helpfulness, but not at test time. Meanwhile, LM-powered multi-agent systems such as Mora [61] and FilmAgent [58] tackle tasks such as scriptwriting and cinematography, yet omit test-time optimization. VISTA combines these directions and is the first to explore black-box prompt optimization for video generation.

3.2. Video and Audio Generation Evaluation

Video generation is uniquely challenging to evaluate: there is no single, definitive “ground truth”, and videos inherently span multiple dimensions that require complex and holistic reasoning to evaluate effectively. Conventional visual quality metrics, such as Inception Score (IS) [50], FID [18], and CLIP-Score [17], typically operate along a single dimension and do not provide a comprehensive evaluation. Recent video benchmarks such as T2I-CompBench [21], VBench [22], and EvalCrafter [33] offer multi-dimensional evaluations tailored to specific model capabilities. However, they still heavily rely on conventional single-metric measures that are time-intensive, inefficient for autonomous improvement, and exclude audio evaluation. Audio-visual extensions like TAVGBench [39] and ACVUBench [60] illustrate complementary: the former emphasizes embedding similarity without failure-focused reasoning, while the latter focuses on evaluating MLLM understanding capabilities. Closest to our MMAC are VideoScore [16] and Evaluation Agent [62], nevertheless, they are not failure-focused and overlook audio dimensions. VISTA emphasizes failure-sensitive visual and audio metrics even for SOTA T2V models, aiming to efficiently enhance AI-generated videos further.

4. Experiments

Benchmarks. We evaluate our approach and baselines on two benchmarks representing distinct scenarios: **single-scene** and **multi-scene** generation. For single-scene evaluation, we follow Dalal et al. [5] to use MovieGenVideo [48] benchmark via randomly selecting 100 prompts. For multi-scene evaluation, we use 161 prompts with at least two scenes from our internal dataset covering diverse topics and matching the T2V model’s duration requirements.

Models and Baselines. VISTA employs two core components: a multimodal large language model

(MLLM) and a text-to-video generation model (T2V). For our experiments, we use **Gemini 2.5 Flash** (Gemini-2.5-flash-preview-05-20) [11] as the MLLM and **Veo 3** (Veo-3.0-generate-preview) [14] as the video generator because they are among the current state of the art. We also experiment with a less powerful T2V model (**Veo 2** [13]) in Section 4.3. Since there is almost no test-time prompt optimization or multi-agent system for video generation up-to-date, we compare VISTA with four baselines: (1) **Direct Prompting (DP)**, which directly uses the user prompt as input for video generation; (2) **Visual Self-Refine (VSR)** [37]², which leverages the MLLM to iteratively evaluate the video generated by the T2V model and subsequently refine the prompt; (3) **Rewrite** [12], which involves using the MLLM to rewrite the user prompt using the Vertex AI video prompt guidelines by Google; (4) **VPO** [4], which expands the user prompts based on three core principles of harmlessness, accuracy, and helpfulness. We run VISTA for 5 iterations: 1 init followed by 4 self-improving steps. In both phases, we sample 5 prompts, each with 3 variants, and generate 2 videos per prompt, resulting in 30 videos per iteration. Since Rewrite and VPO operate in a single iteration, we scale their #videos by matching these used in VISTA’s four self-improvement iterations. Likewise, we scale VSR to match VISTA’s total #videos, denoted as **Visual Self-Refine++ (VSR++)**. For scaled baselines, the best video per iteration is chosen via binary tournament with bidirectional pairwise comparisons (Supp.-Section 8.7).

4.1. Automatic Evaluations

Main Evaluations. We evaluate using both MLLM-as-a-Judge and conventional metrics. Following Dalal et al. [5], we conduct pairwise comparisons between our method and baselines, using **Gemini 2.5 Flash** [11] as the evaluator for its SOTA video-audio understanding and multi-video processing [7, 29]. We assess ten criteria: **Visual Fidelity, Motions, Temporal Consistency (scene level), Text-Video Alignment, Audio Quality, Audio-Video Alignment, Situational Appropriateness, Physical Commonsense, Engagement, and Video Format (beginning, ending, transitions)**. Each comparison is bidirectional, with outcomes recorded as **Win/Tie/Loss** and $\Delta = \text{Win} - \text{Loss}$; conflicts after swapping are marked as Ties. A video wins if it excels in at least three criteria and does not lose on Text-Video Alignment. See Supp.-Section 8.6 for prompts and metric definitions. We also report conventional metrics: **IS** [50], **CLIP-Score** [17], 8 visual metrics from **VBench** [22], and 3 audio metrics from **NISQA** [42]. We also tested other MLLM evaluators including **Gemini 2.5 Pro** [11] and **Qwen2.5-VL-32B-It** [1] in Supp.-Table 9 and found similar trends.

²<https://github.com/madaan/self-refine/blob/main/colabs/Visual-Self-Refine-GPT4V.ipynb>

	Init				2				3				4				5			
Method	Win	Tie	Loss	Δ	Win	Tie	Loss	Δ	Win	Tie	Loss	Δ	Win	Tie	Loss	Δ	Win	Tie	Loss	Δ
Baseline vs. Direct Prompting																				
VSR	22.4	61.2	16.4	6.0	34.9	42.4	22.7	12.2	23.4	48.4	28.1	-4.7	26.6	51.6	21.9	4.7	24.6	47.7	27.7	-3.1
VSR++	28.6†	67.3†	4.1†	24.5†	31.1†	42.2†	26.7†	4.4†	29.5†	52.3†	18.2†	11.3†	26.8†	46.3†	26.8†	0.0†	33.3†	53.3†	13.3†	20.0†
Rewrite	19.0	69.0	12.0	7.0	35.0†	52.0†	13.0†	22.0†	30.0†	56.0†	14.0†	16.0†	39.0†	47.0†	14.0†	25.0†	27.0†	65.0†	8.0†	19.0†
VPO	29.0	46.0	25.0	4.0	34.0†	56.0†	10.0†	24.0†	31.0†	55.0†	14.0†	17.0†	28.0†	65.0†	7.0†	21.0†	36.0†	56.0†	8.0†	28.0†
VISTA	35.5	50.1	14.4	21.1	40.7	49.4	9.9	30.8	41.4	43.7	14.9	26.5	42.4	43.5	14.1	28.3	45.9	50.2	13.9	32.0
VISTA vs. Baselines																				
VSR	47.0	36.4	16.7	30.3	60.0	35.0	5.0	55.0	45.6	43.9	10.5	35.1	38.6	47.4	14.0	24.6	45.6	35.1	19.3	26.3
VSR++	30.4†	51.1†	18.5†	11.9†	48.8†	36.2†	15.0†	33.8†	42.3†	42.3†	15.4†	26.9†	50.7†	29.6†	19.7†	31.0†	30.3†	57.9†	11.8†	18.5†
Rewrite	34.0	51.6	14.4	19.6	34.8†	45.7†	19.6†	15.2†	35.6†	54.4†	10.0†	25.6†	38.2†	42.7†	19.1†	19.1†	40.2†	41.4†	18.4†	21.8†
VPO	27.8	66.7	5.6	22.2	43.8†	50.0†	6.2†	37.6†	40.0†	60.0†	0.0†	40.0†	53.3†	40.0†	6.7†	46.6†	35.7†	45.7†	18.6†	17.1†

(a) Results in single-scene scenarios.

	Init				2				3				4				5			
Method	Win	Tie	Loss	Δ	Win	Tie	Loss	Δ	Win	Tie	Loss	Δ	Win	Tie	Loss	Δ	Win	Tie	Loss	Δ
Baseline vs. Direct Prompting																				
VSR	21.2	63.5	15.4	5.8	33.3	40.4	26.3	7.0	36.7	49.0	14.3	22.4	33.7	52.0	14.3	19.4	35.3	45.9	18.8	16.5
VSR++	29.4†	41.2†	29.4†	0.0†	26.5†	55.9†	17.6†	8.9†	17.6†	58.8†	23.5†	-5.9†	23.5†	52.9†	23.5†	0.0†	26.5†	52.9†	20.6†	5.9†
Rewrite	14.3	55.1	30.6	-16.3	31.8†	55.7†	12.5†	19.3†	29.6†	48.9†	21.6†	8.0†	23.9†	55.7†	20.5†	3.4†	23.9†	52.3†	23.9†	0.0†
VPO	25.2	53.7	21.1	4.1	38.2†	52.8†	9.0†	29.2†	38.2†	51.7†	10.1†	28.1†	28.1†	60.7†	11.2†	16.9†	27.0†	60.7†	12.4†	14.6†
VISTA	37.8	52.3	9.9	27.9	39.4	47.2	13.4	26.0	38.4	52.2	9.4	29.0	43.7	43.7	12.6	31.1	46.3	42.5	11.2	35.1
VISTA vs. Baselines																				
VSR	46.2	44.0	9.9	36.3	43.8	43.8	12.5	31.3	53.2	36.7	10.1	43.1	50.0	40.3	9.7	40.3	48.5	37.9	13.6	34.9
VSR++	35.3†	52.9†	11.8†	23.5†	43.8†	46.9†	9.4†	34.4†	35.0†	46.9†	18.1†	16.9†	34.4†	53.1†	12.5†	21.9†	34.4†	50.0†	15.6†	18.8†
Rewrite	33.5	59.5	8.0	25.5	32.5†	67.5†	0.0†	32.5†	36.6†	57.3†	6.1†	30.5†	37.0†	58.0†	4.9†	32.1†	42.0†	51.9†	6.2†	35.8†
VPO	25.3	64.3	11.4	13.9	27.7†	69.9†	2.4†	25.3†	34.2†	48.8†	17.1†	17.1†	18.5†	74.1†	7.4†	11.1†	25.9†	60.5†	13.6†	12.3†

(b) Results in multi-scene scenarios.

Table 2. Win/Tie/Loss rates and $\Delta = \text{Win} - \text{Loss}$ across 5 iterations. † refers to our scaled-up results, and underlines are results evaluated on half of the benchmark.

Baselines vs. DP. Results. From Table 2, the original baselines (without †) show inconsistent and often unfavorable win-loss dynamics. For example, in Table 2-(a), VPO reaches 29% win rate with only $\Delta = 4.0\%$, while in the multi-scene setting (Table 2-(b)), Rewrite even yields $\Delta = -16.3\%$. This highlights their limited robustness, where gains in some aspects trade off against regressions elsewhere. VSR performs better in multi-scene but adds little in single-scene scenarios, likely because the former is more challenging: DP underperforms in multi-scene, giving VSR room to help, whereas DP’s stronger single-scene performance restricts its impact. Enhanced baselines (with †) mitigate this somewhat by sampling multiple videos and selecting the best, but show no evidence of continued improvement when scaling # videos more. By contrast, VISTA consistently outperforms all baselines and scales more effectively with test-time compute, achieving substantial gains with win rates up to 45.9% ($\Delta = 32\%$) and 46.3% ($\Delta = 35.1\%$).

VISTA vs. Baselines Results. VISTA achieves significant win rates over baselines, ranging from 27.8-60.0% (single-scene) and 18.5-53.2% (multi-scene). Notably, these trends do not correlate with baseline performance relative to DP. For example, while VSR performs best against DP, VISTA surpasses it by the largest margin. This likely reflects

the VISTA’s Pareto optimization of multiple video dimensions, whereas baselines typically focus on fewer aspects with narrower coverage.

Fine-Grained Results. To better understand the metrics that VISTA optimizes, we present a fine-grained pairwise comparison between VISTA and DP in Figure 1. Results reveal that VISTA achieves significant and consistent improvements across key visual, audio, and context dimensions. The most notable gains appear in Visual Fidelity, Engagement, Text-Video Alignment, Motions and Dynamics, Audio Quality, and Situational Appropriateness, attributable to VISTA’s multi-dimensional critique framework and stringent selection process with constraints and penalties. Finally, VISTA yields only marginal improvements in Temporal Consistency and Video Format, as the Veo 3 model already demonstrates strong performance in these aspects.

Conventional Metrics Results. The results on conventional metrics are plotted in Figure 2. While prior frameworks exhibit no clear improvements on both visual and audio criteria over DP, we find that VISTA delivers notable improvements across key visual dimensions, particularly in Dynamic Degree, Aesthetic Quality, and Temporal Style. For

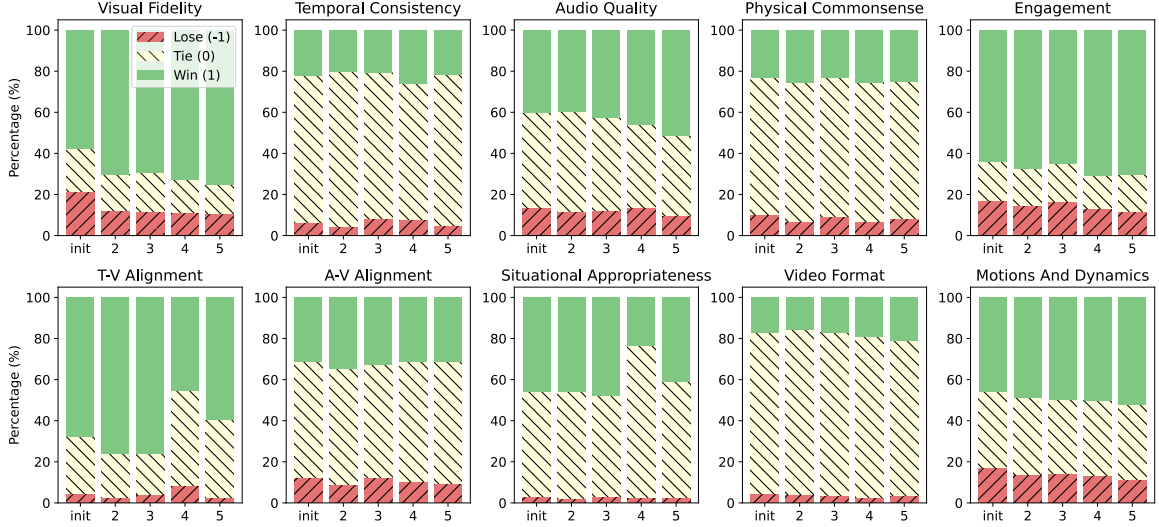


Figure 1. Average win/tie/lose between VISTA DP on two benchmarks. The individual benchmark results are in Supp.-Section 7.5

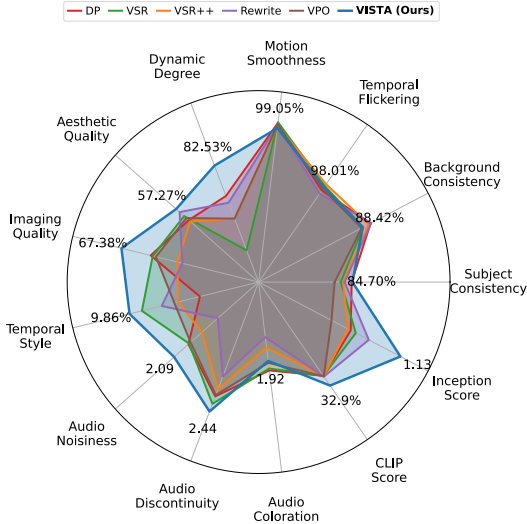


Figure 2. Evaluation results using conventional metrics, averaged over both benchmarks. Scores are rescaled for visualization, and the exact values are provided in Supp.-Tables 4 and 5.

example, it achieves 89.87% in Dynamic Quality in single-scene scenarios, surpassing the best-performing baseline at 77.22%, highlighting the VISTA’s effectiveness in optimizing videos with richer and more realistic motion dynamics. It also reduces audio noisiness (+0.1/5 abs.) and discontinuities (+0.11/5 abs.) across both datasets, and delivers significant gains on CLIP-Score (+3% abs.) versus the best baselines. These gains closely align with the MLLM-as-a-Judge evaluations: both highlight VISTA’s balanced improvements across visual fidelity, audio quality, and alignment.

4.2. Human Evaluations

Main Evaluations. We conduct three human evaluations to assess: (1) if humans prefer VISTA’s outputs over the

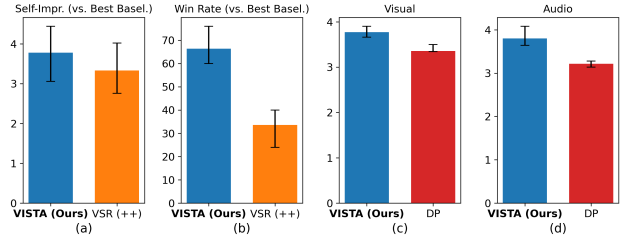


Figure 3. Summary of human evaluation results, with individual scores in Supp.-Table 6

strongest baselines, (2) to what extent VISTA improves video quality across iterations, and (3) to what extent VISTA improves DP on visual and audio quality. For (1), we randomly sample 50 prompts, 25 from single-scene (VISTA vs. VSR++ at 5-th iteration) and 25 from multi-scene scenarios (VISTA vs. VSR at 5-th iteration) and employ five annotators with prompt optimization experience to label each pair as a Win or Loss. For (2), three different expert annotators score those 50 full optimization trajectories of both VISTA and VSR, using a 1-5 scale. For (3), we use the same 50 prompts and ask three expert annotators to rate videos on two specific dimensions: Visual Quality and Audio Quality using a 1-5 scale. See Supp.-Section 9.1 for our instructions. We discuss some corrected cases in Supp.-Section 7.3.

Results. From the human evaluation results shown in Figure 3, VISTA consistently outperforms baselines across all metrics. The win rate comparison in (b) demonstrates its superiority with 66.4% versus 33.6% for best baseline. The self-improvement evaluation in (a) validates that VISTA indeed achieves meaningful self-improvement with an average score of 3.78 out of 5, substantially higher than VSR(++)’s score of 3.33. In addition, the quality assessments in (c) and (d) reveal VISTA’s effectiveness across both visual and audio dimensions. For visual quality in (c), it improves DP

	Single-scene					Multi-scene				
Win Rates over DP	Init	2	3	4	5	Init	2	3	4	5
VISTA	35.5	40.7	41.4	42.4	45.9	37.8	39.4	38.4	43.7	46.3
w/o PROMPTPLANNER	25.2	25.9	30.9	32.9	35.1	34.0	36.1	37.3	37.1	38.8
w/o PAIRWISESELECT	24.5	33.3	29.2	35.4	33.3	27.9	35.5	44.2	38.5	33.8
w/ only Adv. Judge	35.0	40.0	42.0	44.0	42.0	35.3	18.8	18.8	18.8	26.7
w/ only Normal Judge	35.0	32.0	29.3	21.8	17.2	35.3	31.9	37.8	36.6	33.3
w/o DTPA	35.0	36.0	35.6	36.1	37.8	35.3	40.4	40.0	39.5	45.2

Table 3. Ablation results showing distinct contributions of each module in VISTA over two benchmarks.

from 3.36 to 3.77, while for audio in (d), it scores 3.47 versus DP’s 3.21. Finally, while annotators exhibit moderate variability, this is expected due to the inherently subjective nature of videos: annotators tend to prefer different aspects of quality; for example, some place greater weight on visual fidelity, whereas others focus more on detecting awkward physical moments or voices. Nevertheless, all annotators consistently favor the outputs of VISTA over the baselines.

4.3. Analyses

Cost Analysis. We found that VISTA shows an upward trend in performance, ultimately reaching an average of 46.1% win rates as token and video usage increases, with approximately 0.7M tokens and 28 videos consumed per iteration in Supp.-Figure 6. Most token usage comes from tournament selection, with each video evaluation input consuming more than 2K tokens. While these costs are non-trivial, they primarily stem from the current inefficiency of MLLMs, where dense frame-level video input are token-intensive. We view this as a temporary constraint rather than a fundamental limitation: future advances in MLLM compression or specialized evaluators should substantially reduce this overhead. Nonetheless, the steady improvement trend with larger inference budgets indicates that VISTA has strong potential for further test-time scaling.

Ablation Studies. We conduct ablations to assess each component of VISTA: (i) removing PROMPTPLANNER (Step 1) to sample unstructured initial prompts; (ii) removing PAIRWISESELECT (Step 2) to use simple bidirectional comparison; (iii) using only the Adversarial Judge (Step 3); (iv) using only the Normal Judge; and (v) removing DTPA (Step 4) to revise prompts directly from feedback. As shown in Table 3, removing any module degrades performance, confirming that each plays a distinct role. Step 1 removal weakens initialization (Init: 25.2% vs. 35.5% on single-scene; 34.0% vs. 37.8% on multi-scene). Without Step 2, performance becomes unstable occasionally matching other variants early but collapsing later. Using only the Adversarial Judge yields strong single-scene gains yet stagnates on multi-scene (18.8%), while only the Normal Judge quickly collapses (17.2% by iteration 5). Finally, removing DTPA leads to smoother but lower overall improvements, underscoring the need for reasoning-driven prompt revisions in

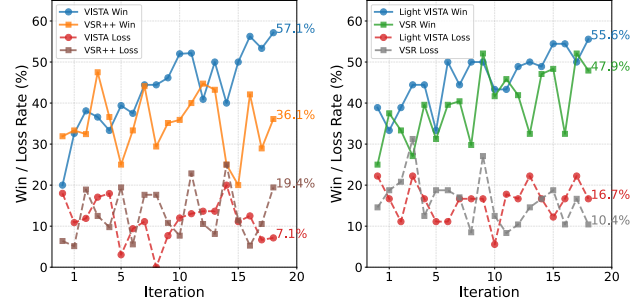


Figure 4. Effect of scaling the #iterations on performance. **Left:** Single-scene. **Right:** Multi-scene.

complex multi-scene generation.

Can VISTA Work with More Iterations? We scale up #iterations in VISTA and compare with the best-performing baselines, as shown in Figure 4. For single-scene scenarios, we compare against VSR++, running both methods for up to 20 iterations with 8 sampled videos per iteration. For multi-scene scenarios, we compare against VSR by running a lightweight version of VISTA (**Light VISTA**) which samples only 1 video per iteration and omits Step 2. We observe the baselines exhibit noisy and inconsistent improvement, whereas VISTA shows a more stable and consistent upward trajectory. These results confirm that our method can be potentially scalable with increased test-time computation.

Can VISTA Work with Other T2V Models? We further test VISTA with Veo 2 [13] and Wan2.2-T2V-A14B [54]. For Veo 2, win rates over DP rise from 15.0% to 23.8% on single-scene and from 27.6% to 33.3% on multi-scene across five iterations. For Wan2.2-T2V-A14B, VISTA achieves high and increasing win rates, from 30.3% to 40.8% on single-scene and from 17.4% to 30.7% on multi-scene across five iterations. Both models show less pronounced gains than Veo 3, likely because they are less capable of fully leveraging the fine-grained details optimized by VISTA. Overall, VISTA effectively enhances generation quality across diverse T2V models.

5. Conclusions

We introduce VISTA, an agentic system designed to self-improve text-to-video generation at test time by jointly optimizing visual, audio, and contextual fidelity through explicit prompt planning, multi-agent critique, and alignment-based video selection. VISTA consistently improves generation quality in SOTA T2V models such as Veo 3, while faithfully preserving user intent and reducing physical, visual, and auditory video hallucinations, leading to substantial human-preference gains. The framework exhibits promising test-time scalability, representing a step toward more adaptive, human-aligned, and self-evolving video generation agents.

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